



Moving around

Navigation

- Right click + drag / Space bar + drag = Panning mode
- + / Mouse wheel = Zoom in
- / Mouse wheel = Zoom out
- + / Space bar = Zoom to selected object
- Ctrl + 1 = 100% zoom
- Ctrl + 0 = Adjust zoom to show all content
- W, e, w = Move on the board
- h, e, H = Shift + 150px = Jump to quick link

Shortcuts to use

Selection

- Click = Single object selection
- Shift + Click = Multi-selection
- Shift + Click & drag = Area selection
- h = Toggle area selection
- Alt + A = Select all
- Alt + T, e, w = Select next element in the direction
- Alt + Shift + + / - Select group child or parent
- Alt = Outlined object and insert to standard element

Tooltips

- A = Add sticky note
- O = Add rectangular sticky note
- S = Add shape
- T = Create text
- A = Add comment
- E = Edit note
- I = Insert into graph
- P = Place/insert/insert in drawing mode
- E = Edit in drawing mode
- Ctrl + L = Lock/unlock object
- Ctrl + Shift + P = Pin/unpin object
- Ctrl + H = Hide/unhide object

Have a play

Use the tools to get familiar with adding content.

The screenshot displays four examples of content created using the 'Have a play' tools. Each example is a square canvas with a grid of colored squares (pink, yellow, green, blue, red) and various text and shape tools. The examples are labeled 'Team 1' through 'Team 4'.

- Team 1:** Features a grid of colored squares. Text elements include 'Bruce', 'Scott', 'Fiona', and 'Hello'. A star shape is also present.
- Team 2:** Features a grid of colored squares. Text elements include 'Ryan' and 'Jackie'. A star shape is also present.
- Team 3:** Features a grid of colored squares. Text elements include 'Claire', 'Kathleen', and 'Paula'. A star shape is also present.
- Team 4:** Features a grid of colored squares. Text elements include 'Kieran' and 'Keith'. A star shape is also present.

Housekeeping

[illegible]

Entity Relationship Diagram

Entity Relationship Diagram showing entities: Actor, Movie, Director, and Movie_Actor. The diagram illustrates the relationships between these entities, including cardinalities and relationship types (e.g., 1:M, 1:1).

Task 1 Write a query to find the top 5 movies based on the number of reviews.

Identify and name potential entities from the information supplied. What are at least 5 entities needed and 2 of these are bridging tables.

1. Director
2. Actor
3. Movie
4. Movie_Director
5. Movie_Actor

[illegible]

Entity Name	Entity Description		
	Action name		Datatype Name
	MemoIR		
Key	Attribute	Entity Type	Comments (Placeholder)
PK	AJL(AR)	INT	Nat Nat
	Action	VecTact(0)	Nat Nat
	Garden	VecTact(1)	Nat

Entity/View type	Where it exists	Where it exists
Database Name	MySQL	
Entity Name	Movie	
PK	Attribute	Data Type
	MovieID	INT
	Year	VarChar(4)
	Rating	INT
	Director	VarChar(64)
	Popularity	Decimal(8,3)
	Release_date	Date
	Season	INT
	Movie_Status	VarChar(16)
	Runtime	INT
	Tagline	VarChar(64)
	Vote_Average	Decimal(3,1)
	Vote_Count	INT

Entity Descriptions	Measure to Characterize Bridge		
Estimation Name	Measured		
Entity Name	Width, Direction		
Key	Direction	Unit Type	Comments (Additional Info)
01	Downhill	ft	Not Used
02	Uphill	ft	Not Used

Entity Description	Match to Actor Bridge		
Database Name	MovieDB		
Entity Name	Movie_Actor		
ID	MovieID	Date Type	Constraint (Null/Not Null)
PK	Actor_ID	INT	Not Null
FK	Movie_ID	INT	Not Null
	Cast_order	INT	Null
	Character_Name	VARCHAR(100)	Null

Task 4
Identify entity relationships. Draw an **Entity Relationship Diagram** showing the relationships (cardinality) between the attributes you described in Task 1.

```
graph LR; Director --> Movie; Director --> Movie; Actor --> Movie_Actor[Movie/Actor]; Movie --> Movie_Actor;
```

[illegible][illegible][illegible]

Entity/Description	Link to the source of all items		
Customer Name	Maria DB		
Service Name	Auto		
City	Atlanta	State Type	Continents (EU/NA/SA)
Primary Key	Auto ID	IS	Not Null
	Auto Name	March 15/20	Not Null
	Gender	March 20/21	IS, F

Entity Description	Entity Type for SPARC			
Entity Name	Character			
Key	Attribute	Entity Type	Constraint (Mandatory)	
	Character	None for (read)	NULL	
Foreign Key	None	Yes	Not Null	
Foreign Key	Yes (Self)	Yes	Not Null	
	Can Delete	Yes	NULL	

[illegible]

Step 1

Identify the entity relationships. These are entity relationship diagrams showing the relationships (constraints) between the attributes (as defined in Step 1)

```
graph LR
    Director1[Director  
Director ID  
Director] --- Producer[Producer  
Director ID  
Movie ID]
    Producer --- Movie1[Movie  
Movie ID  
Director ID  
Producer ID]
    Director1 --- Actor[Actor  
Actor ID  
Actor Name]
    Actor --- Movie2[Movie  
Movie ID  
Director ID  
Producer ID]
    Director1 --- Director2[Director  
Director ID  
Director]
```

Entity Analysis

File Edit Insert View Help

Python 3 (base environment) In [1]:

```
import pandas as pd
import numpy as np
import re
import sys
import os
import glob
import json
import time
import datetime
import logging
import warnings
import itertools
import collections
import random
import string
import hashlib
import pickle
import copy
import math
import statistics
import fractions
import decimal
import fractions
import decimal
import fractions
import decimal
```

In [2]:

```
data = pd.read_csv('data/entities.csv')
data.head()
```

Out[2]:

id	name	type	parent_id	parent_name	parent_type	parent_parent_id	parent_parent_name	parent_parent_type
1	Entity 1	Entity						
2	Entity 2	Entity						
3	Entity 3	Entity						
4	Entity 4	Entity						
5	Entity 5	Entity						

In [3]:

```
data = data[data['type'] == 'Entity']
data = data[data['parent_id'] != None]
```

Out[3]:

id	name	type	parent_id	parent_name	parent_type	parent_parent_id	parent_parent_name	parent_parent_type
1	Entity 1	Entity						
2	Entity 2	Entity						
3	Entity 3	Entity						
4	Entity 4	Entity						
5	Entity 5	Entity						

Entity Analysis

Identify and name potential entities from the information supplied. Here there are at least 5 entities rendered and 2 of these are bridging tables.

1. Director
2. Movie
3. Actor
4. MovieCast
5. MovieDirector

Use the following for your specified tables:

- primary key: PK (indicates the relationship to any other table foreign)
- foreign key: FK (indicates any other Primary and Foreign) (Use Types to search attributes and any available attributes in Null, Not Null)

Entity Descriptions	Director Names		
Instance Name	Movie Database		
Entity Name	Director		
Key	Attribute	Data Type	Constraint (Null/Not Null)
Primary Key	Director ID	Integer	Not Null
	Director	Nvarchar (50)	Not Null

Entity Description	Movies		
Database Name	Movie DB		
	Movie		
	Key	Primary Key	Candidate (Null/Not Null)
PK	MovieID	Int	Not Null
	Tagline	nvarchar(100)	Null
	Title	nvarchar(100)	Not Null
	Budget	Money	Null
	Homepage	nvarchar(250)	Null
	Popularity	Float	Null
	ReleaseDate	Date	Null
	Revenue	money	Null
	Runtime	Int	Not Null
	Movie Data1	nvarchar(50)	Not Null
	VoteAverage	decimal(5,2)	Null
	VoteCount	Int	Null

Entity Description	Actor names		
Database Name	Movie Database		
Entity Name	Actor		
Key	Attribute	Domain	Constraint
PK	Actor_ID	Int	Notnull
	actor_name	nvarchar(100)	Notnull
	Gender	Nvarchar(50)	Notnull

Entity Descriptions	Movie_List		
Database Name	MovieDB		
Entity Name	Movie_List		
Key	movieid		
		Class Type	Constraint (Null/Not Null)
PK	Movie_id	int	Not null
FK	Actor_id	int	Not null
	Character_name	varchar (50)	Not null
	Cast_order	int	Not null

Entity Name	Primary Key	Foreign Key	Relationship
Customer	CustomerID		
Order	OrderID	CustomerID	Customer
Product	ProductID		
Supplier	SupplierID		
Shipment	ShipmentID	OrderID	Order
Inventory	InventoryID	ProductID	Product
Warehouse	WarehouseID		
Location	LocationID		
Region	RegionID		
Country	CountryID		
City	CityID		
State	StateID		
Zip	ZipID		
Address	AddressID		
Phone	PhoneID		
Fax	FaxID		
Email	EmailID		
Website	WebsiteID		
Logo	LogoID		
Image	ImageID		
Document	DocumentID		
Attachment	AttachmentID		
Comment	CommentID		
History	HistoryID		
Log	LogID		
Trace	TraceID		
Event	EventID		
Task	TaskID		
Job	JobID		
Project	ProjectID		
Team	TeamID		
Role	RoleID		
Permission	PermissionID		
Group	GroupID		
Membership	MembershipID		
Session	SessionID		
Connection	ConnectionID		
Database	DatabaseID		
Table	TableID		
Column	ColumnID		
Index	IndexID		
Constraint	ConstraintID		
Trigger	TriggerID		
Procedure	ProcedureID		
Function	FunctionID		
View	ViewID		
Materialized View	Materialized ViewID		
Sequence	SequenceID		
Package	PackageID		
Schema	SchemaID		
Database User	Database UserID		
Database Role	Database RoleID		
Database Group	Database GroupID		
Database Instance	Database InstanceID		
Database Configuration	Database ConfigurationID		
Database Parameter	Database ParameterID		
Database Setting	Database SettingID		
Database Option	Database OptionID		
Database Feature	Database FeatureID		
Database Extension	Database ExtensionID		
Database Add-on	Database Add-onID		
Database Plugin	Database PluginID		
Database Module	Database ModuleID		
Database Component	Database ComponentID		
Database Service	Database ServiceID		
Database Agent	Database AgentID		
Database Task	Database TaskID		
Database Job	Database JobID		
Database Process	Database ProcessID		
Database Thread	Database ThreadID		
Database Connection Pool	Database Connection PoolID		
Database Cache	Database CacheID		
Database Buffer Pool	Database Buffer PoolID		
Database Log	Database LogID		
Database Audit	Database AuditID		
Database Security	Database SecurityID		
Database Encryption	Database EncryptionID		
Database Compression	Database CompressionID		
Database Backup	Database BackupID		
Database Restore	Database RestoreID		
Database Migration	Database MigrationID		
Database Upgrade	Database UpgradeID		
Database Patch	Database PatchID		
Database Patch Set	Database Patch SetID		
Database Patch Group	Database Patch GroupID		
Database Patch Plan	Database Patch PlanID		
Database Patch Task	Database Patch TaskID		
Database Patch Job	Database Patch JobID		
Database Patch Process	Database Patch ProcessID		
Database Patch Thread	Database Patch ThreadID		
Database Patch Connection Pool	Database Patch Connection PoolID		
Database Patch Cache	Database Patch CacheID		
Database Patch Buffer Pool	Database Patch Buffer PoolID		
Database Patch Log	Database Patch LogID		
Database Patch Audit	Database Patch AuditID		
Database Patch Security	Database Patch SecurityID		
Database Patch Encryption	Database Patch EncryptionID		
Database Patch Compression	Database Patch CompressionID		
Database Patch Backup	Database Patch BackupID		
Database Patch Restore	Database Patch RestoreID		
Database Patch Migration	Database Patch MigrationID		
Database Patch Upgrade	Database Patch UpgradeID		
Database Patch Patch	Database Patch PatchID		
Database Patch Patch Set	Database Patch Patch SetID		
Database Patch Patch Group	Database Patch Patch GroupID		
Database Patch Patch Plan	Database Patch Patch PlanID		
Database Patch Patch Task	Database Patch Patch TaskID		
Database Patch Patch Job	Database Patch Patch JobID		
Database Patch Patch Process	Database Patch Patch ProcessID		
Database Patch Patch Thread	Database Patch Patch ThreadID		
Database Patch Patch Connection Pool	Database Patch Patch Connection PoolID		
Database Patch Patch Cache	Database Patch Patch CacheID		
Database Patch Patch Buffer Pool	Database Patch Patch Buffer PoolID		
Database Patch Patch Log	Database Patch Patch LogID		
Database Patch Patch Audit	Database Patch Patch AuditID		
Database Patch Patch Security	Database Patch Patch SecurityID		
Database Patch Patch Encryption	Database Patch Patch EncryptionID		
Database Patch Patch Compression	Database Patch Patch CompressionID		

Task 3
Identify entity relationships. Draw an entity relationship diagram showing the relationship (cardinality) between the entities and the relationships you identified in Task 1.

```
graph TD
    Author[Author] --> Title[Title]
    Title --> Book[Book]
    Book --> Publisher[Publisher]
    Book --> Review[Review]
```